Abstract

This is joint work with Alice C. Niemeyer and Kai-Uwe Schröder.
A topological interlocking assembly (TIA) consists of multiple blocks which are in contact with each other, there exists a subset of the blocks that if it is fixed in place, all other blocks are also fixed in place. Geometries admitting such a TIA are known since 1735, however until recently the number of them was very low. Recent mathematical advances presented a new method of constructing large numbers of candidate geometries which in all investigated cases admitted a TIA. This method is based on the interpolation of deformed fundamental domains of crystallographic groups.

This talk will present the new method as well as a generalisation to the three dimensional case. Additionally, we present results about the mechanical performance of these newly generated TIAs and new approaches to more efficiently evaluate them. Further, we present applications of TIAs from multiple interdisciplinary collaborations, including aerospace, medical devices and biology connected to civil engineering.

As this is ongoing work it might contain preliminary results and present current research hypotheses.

Literature